WAGER: Wealth-Anchored Gain Estimation of Reasoning

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Abstract

On the most label-imbalanced computer-vision benchmarks, a trivial predictor that looks up the most frequent label given cheap context already recovers most of the headline metric, so a reported gain from one method to another may reflect better fitting of annotation-frequency priors rather than genuine visual reasoning. Existing diagnostics are bespoke to a single benchmark and report point estimates with no uncertainty quantification. We introduce *WAGER* (Wealth-Anchored Gain Estimation of Reasoning), a distribution-free, *anytime-valid* framework that decomposes a benchmark gain into a prior-recoverable part and a genuine-reasoning part with a finite-sample validity certificate. The core construction is the *self-prior projection*: rather than comparing a model to an external frequency baseline, we collapse the model's *own* predictions onto the frequency-defining features and treat the result as a frozen competing forecaster. A single testing-by-betting capital process built on this projection simultaneously (i) certifies, via Ville's inequality, that the model uses image information beyond the prior (an e-value), and (ii) quantifies, via its growth rate, the model's usable reasoning information; the measurement is provably invariant to recalibration within a prior cell. Combining two such processes yields the *Reasoning Gain Ratio* (RGR) with an anytime-valid confidence interval and a decision rule (**RGR** $<$ **0.5** flags a gain as non-substantive). On a simulation suite WAGER controls type-I error (0.017 at $\boldsymbol{\alpha}$**=0.05**), attains nominal interval coverage, and recovers an injected reasoning fraction with Pearson $\boldsymbol{r}$**=0.998**. On a real Visual Genome scene-graph-generation study over $\mathbf{7.6 \times 10^5}$ relations, the frequency baseline is correctly certified to perform no reasoning ($\hat{\boldsymbol{I}}_{\mathbf{reason}} \approx \mathbf{0}$, e-value $\approx \mathbf{1}$), and every accuracy gain over it is flagged non-substantive: the bulk of measured "information" is annotation-frequency prior-fitting, not reasoning. WAGER operates entirely on cached probability vectors, so the analysis is cheap and benchmark-agnostic.

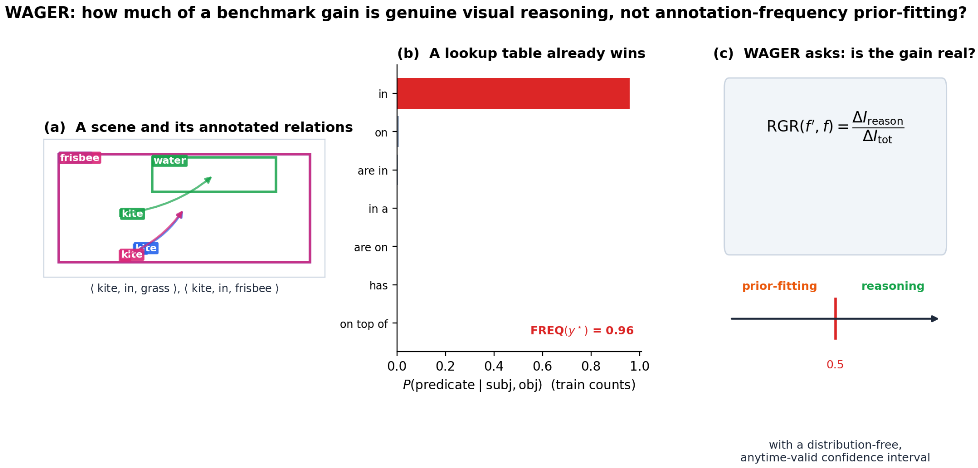

Fig. 1 **The question WAGER answers.** (a) A real Visual Genome scene and its annotated subject–predicate–object relations. (b) For the ordered class pair (subject, object), the training-count distribution $P(\text{predicate} \mid \text{subject}, \text{object})$ is already sharply peaked on the annotated predicate, so a lookup table predicts it *without using a single pixel*. (c) WAGER asks how much of a reported gain from one method to another is genuine reasoning rather than better fitting of this frequency prior, and answers with a Reasoning Gain Ratio carrying a distribution-free, anytime-valid confidence interval.

Keywords: Visual Reasoning, Anytime-Valid Inference, Dataset Bias, E-values, Scene Graph Generation, Visual Question Answering

1 Introduction

1.1 The disease

On the most label-imbalanced vision benchmarks, a trivial predictor that looks up the most frequent label given some cheap context already recovers most of the headline metric:

- **Scene Graph Generation (VG150).** The FREQ baseline of Zellers et al. [1]—predict $\arg\max_p P(\text{predicate} \mid \text{subject}, \text{object})$ from training counts—reaches $R@50 \approx 60.6$ on PredCls versus MOTIFS at 65.2. A lookup table recovers $\sim 96\%$ of "state-of-the-art" recall.
- **Visual Question Answering.** Answering "yes" blindly to "Do you see a..." reaches $\sim 87\%$ on the original unbalanced VQA [2]. On the changing-prior split VQA-CP [3], UpDn collapses from $\sim 63\%$ to $\sim 40\%$. Recent vision-language models remain measurably reliant on such language priors [4, 5].
- **Long-tail detection (LVIS).** Average precision on rare categories is dominated by image-frequency priors [6].

Figure 1 illustrates the mechanism on a real annotation, and Section 4 (Figure 3) shows that on Visual Genome the five most frequent predicates already account for 72% of all relation instances.

1.2 Why existing fixes are not enough

- **mean Recall@K** [7, 8] rebalances metric weight but can be gamed by suppressing head predictions, and it breaks category independence [9].
- **VQA-CP OOD splits** can be solved by *inverting* known label statistics; a random-answer method can beat the state of the art on some question types—a textbook Goodhart failure [10, 11].
- **Causal TDE** [12] and modern debiasing methods [13] are training- or inference-time interventions, not *evaluations* that quantify how much of a reported gain is confound.

Every one of these is bespoke to one benchmark and gives a point estimate with no uncertainty quantification. None answers "is *this specific gain* real, and with what confidence?"

1.3 The gap WAGER fills

We present a general, statistically principled, cross-benchmark test that decomposes a reported gain into a prior-recoverable part and a residual reasoning part, with a finite-sample validity guarantee. Our contributions are:

1. **The self-prior projection** (Definition 1): a new estimator that compares a model to its own frequency-collapsed self, making the null exact without knowing the true prior and the measurement provably calibration-invariant (Corollary 1).
2. **One process, two readouts** (Theorems 1–2): a single testing-by-betting capital process both *certifies* that a model reasons beyond the prior (an anytime-valid e-value) and *quantifies* how much, via its growth rate.
3. **The Reasoning Gain Ratio** (Definition 4, Theorem 3): a gain-attribution statistic with a distribution-free, anytime-valid confidence interval and an operational decision rule.
4. **Validation**: a simulation suite establishing type-I control, nominal coverage, and faithful recovery of injected reasoning, followed by a real Visual Genome study over 7.6×10^5 relations in which the framework certifies the frequency baseline as non-reasoning and flags every accuracy gain over it as non-substantive.

The framework operates entirely on cached probability vectors, so a benchmark is analyzed in minutes on a CPU once logits have been dumped.

2 Related Work

Dataset bias and shortcut learning.

The tendency of models to exploit superficial statistics rather than the intended signal is long documented [14, 15] and was framed as a general failure mode by

Geirhos et al. [16]. The standard responses—reweighting, resampling, or building out-of-distribution splits—change *how models are trained or which test set is used*; they do not deliver a certified measurement of how much of a particular reported gain is shortcut.

Debiasing and evaluation in scene graph generation.

Frequency priors dominate scene graph generation, motivating a large body of debiasing work: re-weighted and re-sampled losses, dynamic tree contexts [8], knowledge routing [7], causal total-direct-effect inference [12], and recent informative-SGG methods [13]. On the evaluation side, mean Recall@K [7, 8] and its independent-mean refinement [9] rebalance metric weight but remain point estimates and can be gamed by suppressing head predictions. WAGER is complementary: it is an *evaluation primitive* that attributes a gain and reports an interval, rather than a method that improves tail recall.

Language priors in VQA and multimodal LLMs.

The "blind" solvability of VQA [2] prompted changing-prior splits [3], which were then shown to be exploitable by label inversion [10, 11]. The problem persists for modern vision-language models: benchmarks such as VLind-Bench [4] and analyses like LanP [5] document substantial reliance on language priors, and a steady stream of debiasing methods continues to appear [17]. These efforts diagnose a benchmark or fix a model; WAGER instead certifies and quantifies a *gain between two systems* with a validity guarantee, and instantiates uniformly across modalities through the choice of prior feature ϕ (Section 4).

Long-tail recognition and detection.

Closely related is long-tailed object detection, where rare-category average precision is dominated by frequency priors [6] and recent work targets the resulting regression and classification bias [18]. WAGER's head/body/tail stratification (Section 4.6) speaks directly to this literature, but measures *usable reasoning per tier* rather than reweighting a metric.

Benchmark reliability and contamination.

A broader reliability crisis now surrounds static benchmarks: surveys and analyses document that leaderboard gains can reflect data contamination and memorization rather than capability, especially for multimodal models where both images and text may leak [19–21]. The community response has largely been *dynamic* or refreshed benchmarks. WAGER is complementary and orthogonal: it leaves the benchmark fixed and instead asks, with a certificate, whether a specific reported gain reflects information beyond a stated confound.

Usable information.

Our reasoning channel measures *usable* (not information-theoretic) information, in the predictive, family-relative sense of $\mathcal{V}$-information [22, 23], conditional probing [24, 25],

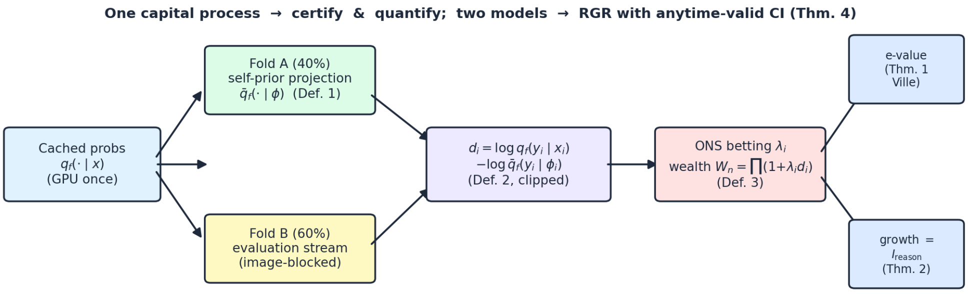

Fig. 2 **The WAGER pipeline.** Cached probabilities are split image-blocked into a projection fold A and an evaluation stream B. The self-prior projection $\bar{q}_f$ is frozen on A (Def. 1); on B the per-instance prior-excess score d_i (Def. 2) drives an Online-Newton-Step betting process whose wealth W_n (Def. 3) is read two ways: as an e-value (Thm. 1) and as a growth rate equal to usable reasoning information (Thm. 2). Two models combine into the RGR with an anytime-valid CI (Thm. 3).

and usable-information dataset diagnostics [26]. WAGER's novelty is to measure this quantity *beyond a self-induced prior baseline* and to attach an anytime-valid certificate to it.

Game-theoretic statistics and anytime-valid inference.

WAGER is built on testing-by-betting and e-values [27–30], the bounded-mean capital process of Waudby-Smith and Ramdas [31], and the forecaster-comparison e-process of Henzi and Ziegel [32]; the area remains highly active, with recent work sharpening the foundations and the methodology [33–35] and extending e-values to applied sequential monitoring [36]. The growth-rate–information identity goes back to Kelly [37] and the universal-portfolio view [38, 39]. To our knowledge WAGER is the first to cast benchmark-confound attribution as a game-theoretic e-process, yielding a single object that both certifies (Theorem 1) and quantifies (Theorem 2) reasoning beyond annotation-frequency priors. Unlike conditional-independence testing, which is provably hard in general [40], we run an honest predictive comparison of two fixed forecasters, which is exactly what betting certifies.

3 Method

3.1 Problem setup and notation

Let x be an instance (for scene graph generation, an ordered object pair and its image region), $y \in \{1, \dots, K\}$ the ground-truth label, and $\phi(x)$ the *prior features*: the cheap, frequency-carrying sufficient statistic through which annotation frequency acts (for SGG, $\phi(x) = (\text{subject class}, \text{object class})$). A model f supplies a predictive distribution $q_f(y \mid x)$ (a softmax of cached logits); $u(y) = 1/K$ is the uniform reference. We never assume q_f is calibrated, nor that the true conditionals $p(y \mid x)$ or $p(y \mid \phi)$ are known. We assume only that instances on an evaluation stream B are *exchangeable*, which we enforce by randomizing order with image-blocked permutations (Section 4).

3.2 The self-prior projection

Definition 1 (Self-prior projection)**.** For a fixed model f, its self-prior projection is the model's own predictions averaged within each prior-feature cell,

$$\bar{q}_f(y \mid \phi) \;=\; \mathbb{E}[\, q_f(y \mid X) \mid \phi(X) = \phi \,]\,. \tag{1}$$

By construction $\bar{q}_f$ is the best summary of f measurable with respect to the prior features alone: a forecaster that has discarded every pixel-level cue and kept only what frequency context implies. The crux is that we compare f to its *own* frequency-collapsed self rather than to an external FREQ table whose mismatch with the true prior would break validity; the two share calibration, vocabulary, and idiosyncrasies, so any out-of-fold predictive advantage of q_f over $\bar{q}_f$ can come only from information beyond ϕ.

Estimation on real, heavy-tailed data.

We estimate $\bar{q}_f$ on a fold A disjoint from the evaluation stream. The naive cell mean is unusable when most cells are sparse (Figure 3), so we use a hierarchical empirical-Bayes backoff with Krichevsky–Trofimov-style smoothing. Writing n_ϕ for the cell count and $\pi(\phi)$ for a parent distribution,

$$\hat{\bar{q}}_f(\cdot \mid \phi) \;=\; \frac{\sum_{j \in A} q_f(\cdot \mid x_j) \nVdash [\phi(x_j) = \phi] \;+\; m\, \pi(\phi)}{n_\phi + m}, \tag{2}$$

with the backoff hierarchy cell $(\mathrm{subj}, \mathrm{obj}) \to$ coarse $(\mathrm{subj}) \to$ global mean. A second-order Jensen correction, computed from the within-cell variance, debiases the log-projection; it cancels in the total information I^{tot} and only re-allocates mis-credited reasoning back into the prior channel. A model whose predictions are a function of ϕ alone (FREQ, a class-only network) uses the identical backoff, so its out-of-fold reasoning score collapses to ≈ 0—this is exactly what makes the FREQ anchor hold on real data (Section 4).

3.3 Per-instance score and the wealth process

Definition 2 (Prior-excess log-score)**.** On the evaluation stream, define

$$d_i \;=\; \log q_f(y_i \mid x_i) - \log \hat{\bar{q}}_f(y_i \mid \phi(x_i)), \qquad \tilde{d}_i = \mathrm{clip}(d_i, -c, c), \tag{3}$$

with default clip $c = \log K$. Here $d_i > 0$ means that at instance i, using the image beyond ϕ helped predict the true label.

Definition 3 (Wealth process). We test $H_0: \ \mu := \mathbb{E}[\tilde{d}_i] \le 0$ ("f predicts no better than its prior-projection") against $H_1 : \mu > 0$ via

$$W_0 = 1, \qquad W_n = \prod_{i=1}^{n}\bigl(1 + \lambda_i\, \tilde{d}_i\bigr), \qquad \lambda_i \in [0, \tfrac{1}{c}) \text{ predictable}, \tag{4}$$

with λ_i chosen online by the Online-Newton-Step (aGRAPA) rule of Waudby-Smith and Ramdas [31], a closed-form predictable update requiring no tuning.

Theorem 1 (Anytime-valid certificate). *Under H_0, (W_n) is a non-negative supermartingale with $\mathbb{E}[W_n] \le 1$; it is therefore an e-value, and by Ville's inequality [41], for any $\alpha \in (0, 1)$, $\Pr[\exists n : W_n \ge 1/\alpha] \le \alpha$. Rejecting H_0 the first time $W_n \ge 1/\alpha$ controls type-I error at level α with no fixed sample size and no distributional assumption beyond exchangeability.*

Proof sketch λ_i is $\mathcal{F}_{i-1}$-measurable and $1 + \lambda_i\tilde{d}_i \ge 0$ because $\lambda_i < 1/c$ and $\tilde{d}_i \ge -c$. Under H_0, $\mathbb{E}[1 + \lambda_i\tilde{d}_i \mid \mathcal{F}_{i-1}] = 1 + \lambda_i\, \mathbb{E}[\tilde{d}_i \mid \mathcal{F}_{i-1}] \le 1$, so (W_n) is a non-negative supermartingale; telescoping gives $\mathbb{E}[W_n] \le 1$, and Ville's inequality yields the bound. The design choice that buys validity *without knowing the true prior* is that $\bar{q}_f$ is frozen on fold A and treated purely as a competing forecaster: we run an honest predictive comparison of two fixed forecasters, not a conditional-independence test (which is hard in general [40]). □

Theorem 2 (Growth rate = usable reasoning information). *With the log-optimal predictable bet $\lambda_i^\star = \arg\max_\lambda \mathbb{E}[\log(1 + \lambda\tilde{d}) \mid \mathcal{F}_{i-1}]$, the growth rate converges almost surely, and as $c \to \infty$,*

$$G(f) := \lim_{n\to\infty} \tfrac{1}{n} \log W_n = \mathbb{E}[\log q_f(Y \mid X) - \log \bar{q}_f(Y \mid \phi(X))] =: \mathcal{I}_f^{\text{reason}} \ \ge 0. \tag{5}$$

If q_f equals the true posterior and $\bar{q}_f$ the true $p(\cdot \mid \phi)$, then $\mathcal{I}_f^{\text{reason}} = I(Y; X \mid \phi(X))$, the conditional mutual information.

Proof sketch By Kelly [37] and Cover and Thomas [39], the log-optimal growth rate of a forecasting game equals the expected log-score differential, which the strong law drives to $\mathbb{E}[\tilde{d}]$; as $c \to \infty$, $\tilde{d}_i \to d_i$. The chain rule $I(Y; X) = I(Y; \phi(X)) + I(Y; X \mid \phi(X))$, valid because ϕ is a function of X, identifies the residual with conditional mutual information in the calibrated case. □

The same wealth process thus *certifies* (via W_n) and *quantifies* (via $G(f)$) reasoning beyond the prior—the central unification.

Corollary 1 (Calibration invariance). *$\mathcal{I}_f^{\text{reason}}$ is invariant to any recalibration map applied uniformly within a ϕ-cell (e.g. per-cell temperature scaling), because the map cancels between q_f and $\bar{q}_f$. Hence miscalibration shared within a prior cell cannot inflate measured reasoning; by Gibbs' inequality an arbitrarily miscalibrated q_f can only* under-*estimate the true reasoning information, never manufacture it.*

3.4 The Reasoning Gain Ratio

Decompose each model's total information over uniform into a prior-fitting and a reasoning channel,

$$\underbrace{\mathbb{E}[\log \bar{q}_f(Y \mid \phi) - \log u(Y)]}_{I_f^{\text{prior}}} + \underbrace{\mathbb{E}[\log q_f(Y \mid X) - \log \bar{q}_f(Y \mid \phi)]}_{I_f^{\text{reason}}} = I_f^{\text{tot}}. \tag{6}$$

Definition 4 (Reasoning Gain Ratio)**.** For a claimed improvement from f to f',

$$\text{RGR}(f', f) = \frac{I_{f'}^{\text{reason}} - I_f^{\text{reason}}}{I_{f'}^{\text{tot}} - I_f^{\text{tot}}}, \tag{7}$$

the fraction of the information gain attributable to better reasoning rather than better prior-fitting. **Decision rule:** if the upper confidence limit of RGR is below 0.5, the majority of the gain is prior-recoverable and the claim is flagged *non-substantive*.

Theorem 3 (Anytime-valid CI for RGR)**.** *Each of I^{reason} and I^{tot} is the mean of a bounded score, so the betting confidence sequence of [31] gives each an anytime-valid $(1-\alpha)$ interval. Combining numerator and denominator intervals by Fieller's method [42] under a Bonferroni split yields a $(1-2\alpha)$ anytime-valid interval for $\text{RGR}(f', f)$.*

3.5 Algorithm and implementation

Algorithm 1 summarizes the driver. Two practical points make it cheap and exact. First, the GPU is used *once*, only to dump cached probability vectors; all WAGER computation is CPU NumPy with an $O(NK)$ hot loop. Second, because the per-instance scores are order-independent under a frozen train-fit projection, the R image-blocked permutations only reshuffle the betting order to build the e-value distribution and average the anytime-valid intervals across orders.

3.6 Design rationale

Four choices distinguish WAGER from a naive "compare to FREQ" diagnostic, and each is what makes the certificate valid rather than merely suggestive.

Self-prior, not an external baseline.

If we compared q_f to an external frequency table $\hat{p}(y \mid \phi)$ built from training counts, the null "f reasons no better than the prior" would only hold if that table equaled the true $p(y \mid \phi)$—which it never exactly does. The mismatch is a free parameter that an adversary (or an unlucky dataset) can exploit to manufacture or hide reasoning. Freezing $\bar{q}_f$ from f's own predictions removes that parameter: under H_0 the comparison is between two forecasters that share every nuisance (calibration, vocabulary bias, temperature), so any out-of-fold edge is attributable to ϕ-excess information alone.

Algorithm 1 WAGER core routine

Require: cached q_f for each model; prior map ϕ; coarse key; level α; clip $c = \log K$; permutations R.
Ensure: per-model e-value and $\hat{I}^{\text{reason}}, \hat{I}^{\text{prior}}$ with CIs; per-pair RGR with verdict.

fit frozen $\bar{q}_f$ on fold A via hierarchical backoff (Def. 1)
for each model f and each i in stream B **do**
  $d_i \leftarrow \text{clip}(\log q_f(y_i \mid x_i) - \log \bar{q}_f(y_i \mid \phi_i), -c, c)$; $e_i \leftarrow \text{clip}(\log \bar{q}_f(y_i \mid \phi_i) - \log u(y_i), -c, c)$
end for
for $r = 1 \dots R$ (image-blocked permutations of B) **do**
  $\lambda_i \leftarrow \text{ONS}(\tilde{d}_{1:i-1})$; $W_n \leftarrow \prod(1 + \lambda_i d_i)$ ▷ e-value, growth (Thm. 1,2)
end for
$\hat{I}^{\text{reason}} \leftarrow \text{mean}(d)$, $\hat{I}^{\text{prior}} \leftarrow \text{mean}(e)$, with betting CIs
for each pair: RGR $\leftarrow \Delta\hat{I}^{\text{reason}}/\Delta\hat{I}^{\text{tot}}$ with Fieller CI (Thm. 3); verdict by the 0.5 rule

Betting, not a conditional-independence test.

Asking "is $Y \perp X \mid \phi(X)$ under f?" is a conditional-independence question, and consistent CI testing without further assumptions is impossible [40]. WAGER sidesteps this by testing a *predictive* mean ($\mathbb{E}[\tilde{d}] \le 0$) with a betting martingale, which is exactly the regime where game-theoretic statistics gives finite-sample validity [27, 29, 31].

Clipping for boundedness.

The wealth factor $1 + \lambda\tilde{d}$ must stay non-negative for the supermartingale argument; clipping d to $[-c, c]$ and capping $\lambda < 1/c$ guarantees it. The clip is the one knob that trades robustness against sensitivity (Section 4.8); we fix $c = \log K$, the natural scale of a log-likelihood ratio over K classes.

Image-blocked exchangeability.

Multiple relations share an image and are not independent. Permuting at the image level (all pairs of an image move together, in randomized image order) yields an exchangeable stream at the block level, which is all Theorem 1 needs, and averaging over R such permutations stabilizes the e-value and the interval.

3.7 Computational cost

WAGER consumes cached probability vectors, never models. The GPU is used once, to dump $\{(y_i, q_f(\cdot \mid x_i), \phi_i)\}$ to disk; all subsequent computation is CPU NumPy. Fitting the projection is a single pass of cell-wise accumulation ($O(N_A K)$); the betting stream and the prior channel are $O(N_B K)$; the betting confidence sequence evaluates a grid of G candidate means with a vectorized fixed-λ hedged capital process ($O(N_B G)$ but fully vectorized and chunked to bound memory). The full five-model VG study, including R permutations and all intervals, completes in roughly three minutes on a single CPU core after the one-time logit dump, which is why the framework scales to large test sets and many model pairs.

3.8 A worked micro-example

To make the score concrete, consider a cell $\phi = (\text{man}, \text{horse})$ with frozen projection $\bar{q}_f(\text{riding} \mid \phi) = 0.30$. For an instance whose true predicate is "riding" and where the model, seeing the boxes overlap with the man above the horse, outputs $q_f(\text{riding} \mid x) = 0.62$, the reasoning score is $d = \log 0.62 - \log 0.30 = +0.73 > 0$: the image moved the prediction toward the truth beyond what the class pair alone implies, and the bettor profits. For a second instance where the model returns the cell-typical 0.30, $d = 0$ and the bettor neither gains nor loses. A model that is merely *confident* but constant within the cell (e.g. always outputting 0.30) produces $d \equiv 0$ and earns no capital—this is the mechanism behind the FREQ anchor and behind calibration-invariance (Corollary 1).

4 Experiments

We first describe the corpus (Section 4.1), validate WAGER on a simulation suite where ground truth is known (Section 4.3), apply it to a real Visual Genome scene-graph study (Sections 4.4–4.6), and finally probe sensitivity to design choices (Section 4.8). All reported intervals are at $\alpha = 0.05$ unless stated.

4.1 Dataset and statistics

Our study uses the Visual Genome scene-graph corpus [43] in the PredCls regime (ground-truth boxes and object labels given, predicate to be predicted), with the standard VG150 vocabulary of 150 object classes and 50 predicates. Table 1 summarizes the corpus. Two properties make it an ideal stress test for WAGER. First, it is *heavy-tailed*: the five most frequent predicates account for 71.6% of all relation instances and the single predicate "on" for 32.4% (Figure 3), so a frequency lookup already secures most of any recall-style metric. Second, the prior-feature space is *sparse*: only 49.2% of the 150×150 possible ordered class pairs are observed in training, and 46.4% of the observed cells contain fewer than five instances—which is exactly why a naive cell mean fails and the hierarchical backoff of Section 3 is required.

4.2 Prior features and benchmark generality

WAGER is instantiated through the prior feature ϕ, the statistic through which annotation frequency acts (Table 2). Our empirical study targets Visual Genome PredCls, where ϕ is the ordered class pair; the same machinery applies to VQA/GQA (question type) and LVIS (frequency tier) without modification, which is the sense in which WAGER is benchmark-agnostic.

4.3 Simulation validity and power

Before touching real models we verify the framework's guarantees on synthetic data with known ground truth, by mixing a prior forecaster with an oracle, $q_f = (1 - \beta)\,\text{prior} + \beta\,\text{oracle}$, so the true reasoning fraction is controlled by β. Figure 4 summarizes three gates, all of which pass:

Table 1 Visual Genome PredCls corpus statistics. The test images are split image-blocked into a projection fold (for $\bar{q}_f$) and an evaluation stream (for betting).

Quantity	Value
Relation instances (total)	762,592
training / test	532,987 / 229,605
test: projection / eval fold	113,255 / 116,350
Images (train / test)	65,228 / 27,955
Relations per image (mean / median, test)	8.21 / 6
Predicate classes K	50
Object classes	150
Prior-feature cells (observed / possible)	11,076 / 22,500
cells with <5 training instances	46.4%
Top-5 predicate share / "on" share	71.6% / 32.4%
Head / body / tail predicates (#)	5 / 11 / 34
Head / body / tail instance share	71.6% / 16.6% / 11.8%

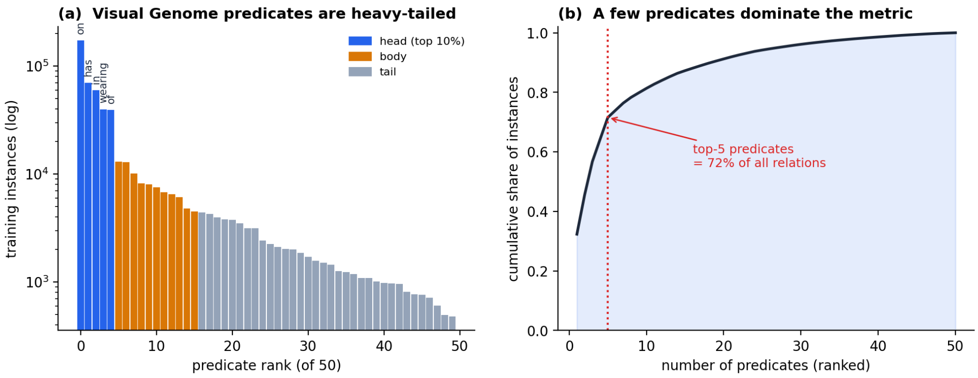

Fig. 3 **The confound is severe.** (a) Visual Genome's 50 predicates follow a heavy-tailed training distribution (log scale; head/body/tail). (b) The five most frequent predicates account for 72% of all relation instances, so a frequency lookup already secures most of the recall metric.

- **Type-I error.** Feeding $q_f = \bar{q}_f$ (model = prior) over 600 runs, the empirical exceedance rate is $0.017 \leq \alpha = 0.05$, and $\hat{I}^{\text{reason}} \approx 0.004$, confirming Theorem 1.
- **Coverage.** The anytime-valid intervals attain 0.93 empirical coverage against a 0.90 target.
- **Recovery.** The estimated RGR tracks the injected ground-truth RGR with Pearson $r = 0.998$ and is monotone in β; $\hat{I}^{\text{reason}}$ tracks the true reasoning information with $r = 0.999$.

Table 2 Prior feature ϕ per benchmark. The empirical study in this paper uses the first row.

Benchmark	Task	Prior feature $\phi(x)$
Visual Genome (VG150)	Scene graph generation	(subject class, object class)
GQA / VQA v2	Visual question answering	question type / template
LVIS	Long-tail detection	category-frequency tier

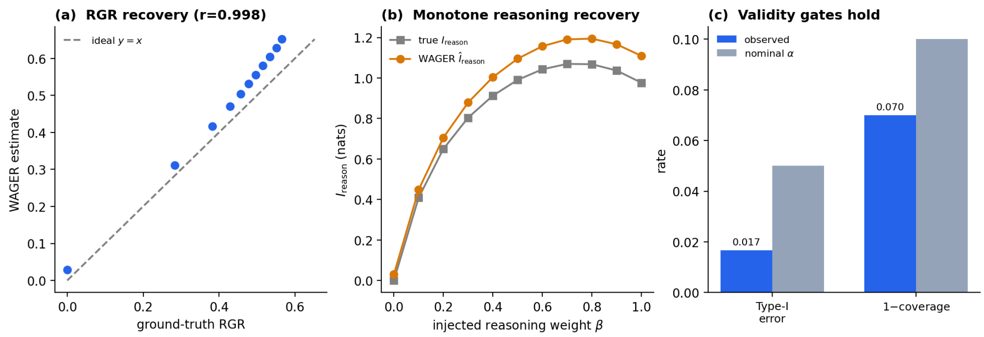

Fig. 4 **Phase-1 validation.** (a) Estimated vs. ground-truth RGR ($r = 0.998$). (b) Monotone recovery of injected reasoning information. (c) Type-I error and one-minus-coverage stay within their nominal levels.

4.4 Real study: models and the anchor

We build five predicate predictors of increasing access to image information beyond the class pair: **FREQ** (training-count lookup $P(\text{predicate} \mid \text{subject}, \text{object})$, constant within a ϕ-cell); **FREQ+OVERLAP** (FREQ plus a binary box-overlap indicator); **MLP-CLASS** (an MLP on class embeddings only—still essentially ϕ-measurable); **MLP-SPATIAL** (adds translation/scale-invariant box geometry: relative position, scale, aspect, IoU); and **MLP-SPATIAL+** (a larger spatial network). The spatial features are the "pixels beyond ϕ" that distinguish reasoning from prior-fitting. All five are trained once; WAGER then consumes their cached predicate distributions.

The anchor holds.

Table 3 reports the readouts. As the credibility gate demands, FREQ returns $\hat{I}^{\text{reason}} = -0.005$ (CI straddling/below 0) and e-value $0.49 \approx 1$: WAGER correctly certifies that a lookup table performs no visual reasoning. FREQ+OVERLAP is likewise certified non-reasoning. The MLPs that genuinely consult box geometry accumulate decisive evidence (e-values up to the numerical ceiling $\sim 10^{304}$) and positive $\hat{I}^{\text{reason}}$, growing with spatial capacity. Crucially, test accuracy moves by under 1.3 points across all five models, so accuracy alone cannot tell whether a gain is reasoning or prior-fitting—the distinction WAGER is built to make.

Table 3 WAGER readouts on Visual Genome PredCls (eval $N = 116{,}350$; $\alpha = 0.05$). I^{reason} shown with its anytime-valid CI. Test accuracy moves by under 1.3 points across all five models, yet the e-value and reasoning channel separate them sharply.

Model	e-value	$\hat{I}^{\text{reason}}$ (CI)	I^{prior}	I^{tot}	acc.
FREQ	0.49	$-0.005\,[-0.006, -0.001]$	2.524	2.518	0.6564
FREQ+OVERLAP	0.12	$-0.017\,[-0.018, -0.011]$	2.477	2.460	0.6545
MLP-CLASS	$4{\times}10^{145}$	$+0.028\,[+0.024, +0.030]$	2.688	2.716	0.6557
MLP-SPATIAL	$> 10^{300}$	$+0.063\,[+0.056, +0.066]$	2.699	2.762	0.6639
MLP-SPATIAL+	$> 10^{300}$	$+0.071\,[+0.063, +0.074]$	2.710	2.781	0.6674

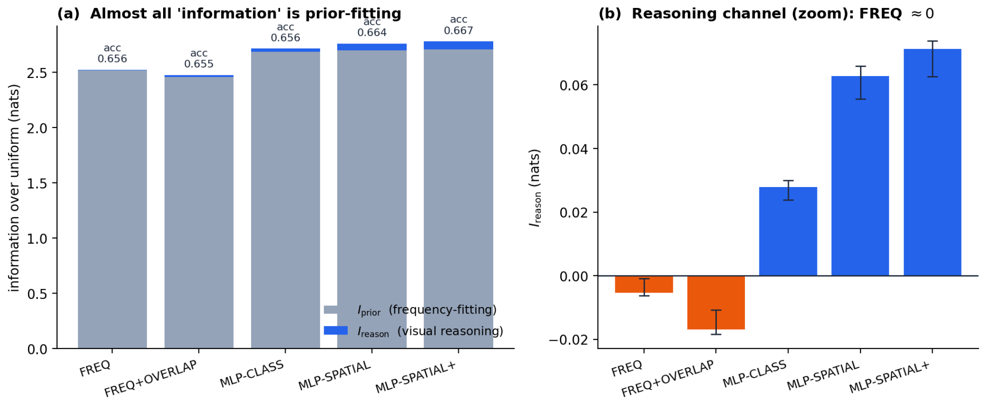

Fig. 5 **Information decomposition.** (a) Stacked I^{prior} (slate) and I^{reason} (blue) per model with test accuracy; the reasoning sliver is barely visible against prior-fitting. (b) The reasoning channel alone, with anytime-valid CIs: FREQ/FREQ+OVERLAP are ≤ 0, the spatial MLPs are clearly positive.

Almost all "information" is prior-fitting.

Figure 5 decomposes total information into the two channels. For every model the prior channel ($I^{\text{prior}} \approx 2.5$ nats) dwarfs the reasoning channel ($I^{\text{reason}} \leq 0.07$ nats): even the strongest model spends over 97% of its information budget on frequency-fitting. Figure 6 shows the testing-by-betting view directly: the capital of FREQ and FREQ+OVERLAP never leaves the neighborhood of its starting value, while the spatial MLPs compound capital linearly in log-scale—visual confirmation that only those models out-predict their own self-prior projection. Figure 7 illustrates the mechanism inside a single ϕ-cell: the projection $\bar{q}_f$ is the frozen cell forecaster, and the spread of per-instance scores d_i around it is exactly what the betting process harvests.

4.5 Qualitative examples

Figure 8 grounds the per-instance score on the actual images we process. For each annotated relation we show the predicted predicate, the FREQ top-1 prediction, and

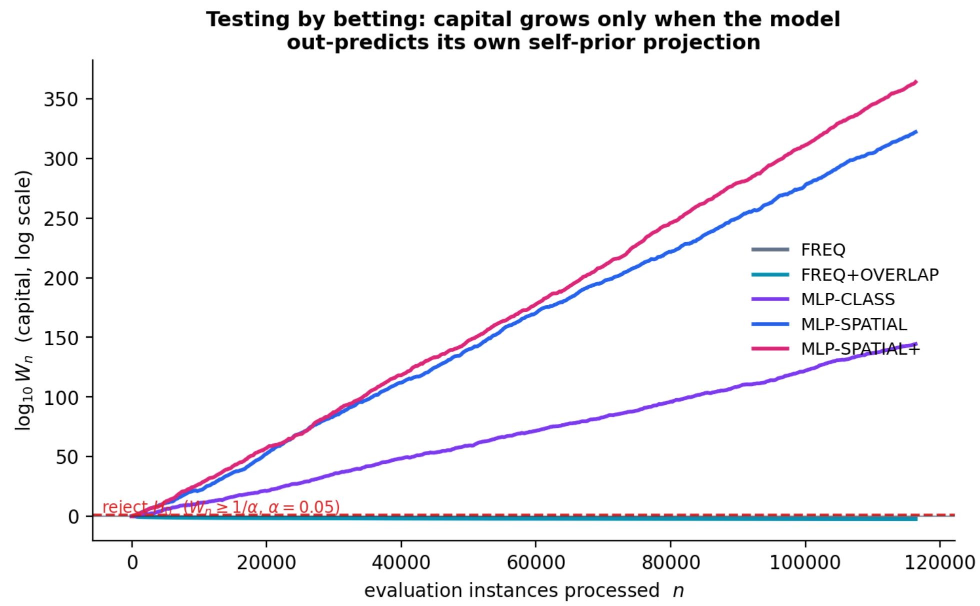

Fig. 6 Testing by betting. Log-capital $\log_{10} W_n$ over the evaluation stream. FREQ and FREQ+OVERLAP stay flat (no evidence of reasoning); the spatial MLPs compound capital and cross the rejection threshold $1/\alpha$ almost immediately.

WAGER's score d_i for MLP-SPATIAL. Where the geometry is informative—a relation whose predicate depends on the relative position or overlap of the two boxes—d_i is clearly positive ("image helps"); where the class pair alone already determines the likely predicate, $d_i \approx 0$ ("prior suffices"). This is the instance-level view of the aggregate decomposition: WAGER does not reward a confident prediction per se, only one that beats the model's own frequency-collapsed self.

4.6 Where reasoning lives: head, body, and tail

An aggregate I^{reason} can hide *where* a model reasons. We stratify the per-instance scores by predicate frequency tier (head: top 5 predicates; body: next 11; tail: remaining 34) and report the mean reasoning score $\bar{d}$ per tier in Figure 9 and Table 4. The pattern is striking and consistent: *head predicates are essentially won by the prior* (for the spatial MLPs $\bar{d} \leq 0.026$, and for FREQ it is even slightly negative), whereas the *body and tail carry almost all of the reasoning signal* ($\bar{d} = 0.16$–0.19 for MLP-SPATIAL/+). In other words, the small aggregate reasoning channel is not spread uniformly; it is concentrated on exactly the rare predicates that headline recall metrics underweight. This is the WAGER-native restatement of a familiar concern—that aggregate scores mask tail behavior [9, 18]—now with a distribution-free per-tier measurement rather than a re-weighted metric.

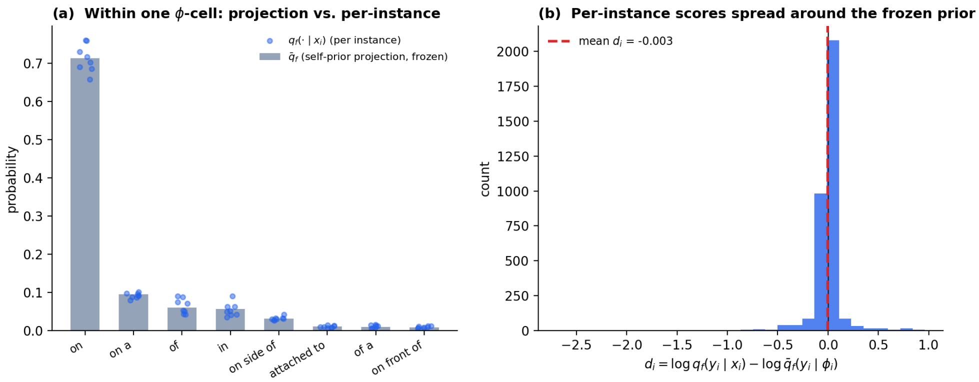

Fig. 7 **Self-prior projection in one ϕ-cell.** (a) The frozen projection $\bar{q}_f$ (bars) versus per-instance model predictions $q_f(\cdot \mid x_i)$ (dots). (b) The distribution of per-instance reasoning scores d_i: their spread around the prior is the signal WAGER measures.

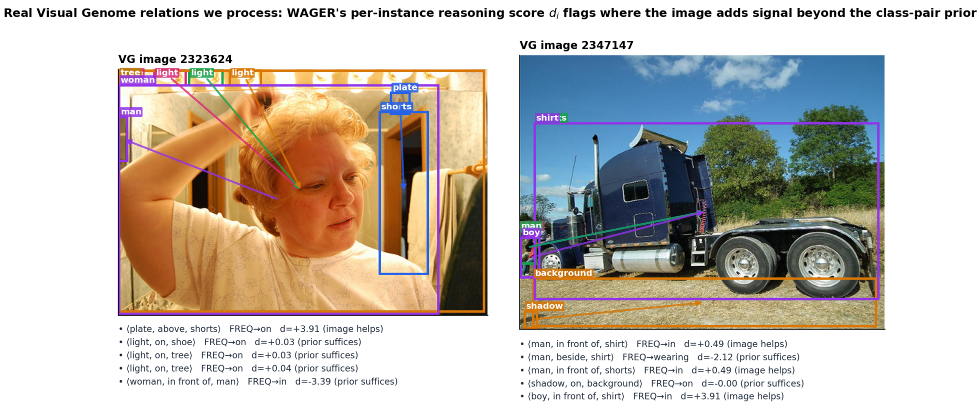

Fig. 8 **Real processed examples.** Visual Genome images with our overlaid subject/object boxes and relation arrows. Each caption lists the annotated triplet, the FREQ top-1 guess, and the per-instance reasoning score d_i for MLP-SPATIAL: positive d_i marks relations where the image adds signal beyond the class-pair prior.

4.7 Headline finding: the gains are non-substantive

Figure 10 and Table 5 give the gain attribution. Every accuracy gain over the frequency baseline is flagged *non-substantive*: the upper CI of its RGR is well below 0.5, so most of each reported improvement is prior-recoverable rather than reasoning. The decision rule is not vacuous—it fires positively when warranted: the gain from MLP-CLASS to MLP-SPATIAL, which adds genuine box geometry, is *reasoning-supported* (RGR $=$ 0.77, CI $[0.74, 0.79]$). The further step to MLP-SPATIAL+ adds capacity but little new visual signal and is again non-substantive (RGR $=$ 0.44). WAGER thus distinguishes a gain that comes from new image information from one that comes from

Table 4 Mean per-instance reasoning score $\bar{d}$ (nats) by predicate-frequency tier. FREQ stays near zero in every tier; the spatial MLPs concentrate their reasoning on body and tail predicates.

Model	head	body	tail
FREQ	−0.021	+0.033	+0.038
FREQ+OVERLAP	−0.032	+0.025	+0.017
MLP-CLASS	+0.014	+0.069	+0.057
MLP-SPATIAL	+0.024	+0.162	+0.157
MLP-SPATIAL+	+0.026	+0.189	+0.182

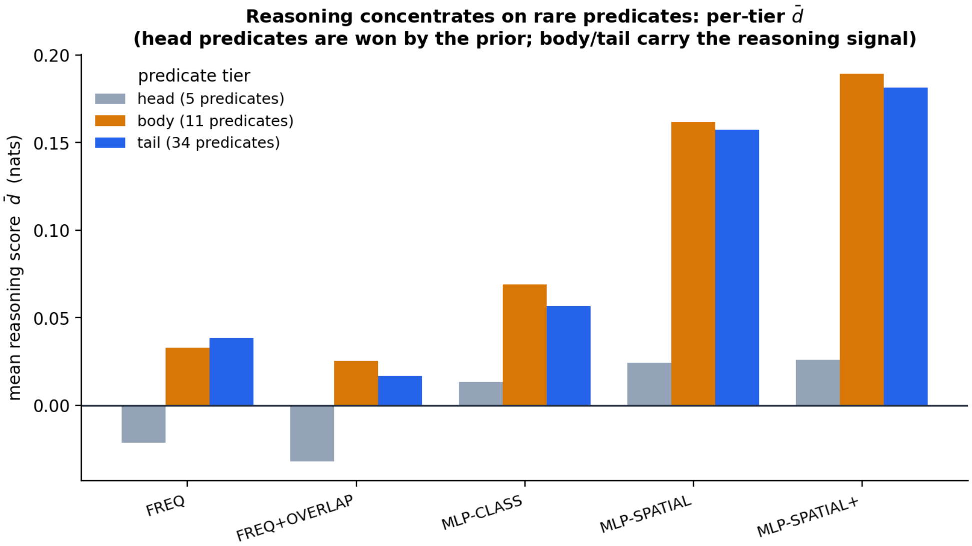

Fig. 9 **Reasoning concentrates on rare predicates.** Mean reasoning score $\bar{d}$ per predicate tier. Head predicates are won by the prior; the body and tail carry the reasoning signal, and it grows with spatial capacity.

better prior-fitting or extra capacity, which raw accuracy (spread of under 1.3 points across all five models) cannot.

4.8 Ablations

Figure 11 varies the main design choices for the MLP-SPATIAL-vs-FREQ gain. The verdict is stable across the prior-feature granularity (cell vs. subject-only), the shrinkage pseudo-count $m \in \{0, 0.25, 1\}$, and the number of permutations $R \in \{10, 30, 60\}$: all remain non-substantive. The one sensitive knob is the clip c: at the most aggressive clipping ($c = \log K/2$) the ratio crosses 0.5, because clipping compresses the large

Table 5 Reasoning Gain Ratio with Fieller CI and verdict on VG PredCls.

Gain f' vs f	RGR (CI)	Verdict
FREQ+OVERLAP vs FREQ	0.20 $[0.17, 0.22]$	non-substantive
MLP-CLASS vs FREQ	0.17 $[0.16, 0.18]$	non-substantive
MLP-SPATIAL vs FREQ	0.28 $[0.27, 0.29]$	non-substantive
MLP-SPATIAL+ vs FREQ	0.29 $[0.28, 0.30]$	non-substantive
MLP-SPATIAL+ vs MLP-SPATIAL	0.44 $[0.39, 0.49]$	non-substantive
MLP-SPATIAL vs MLP-CLASS	**0.77** $[\mathbf{0.74}, \mathbf{0.79}]$	**reasoning-supported**

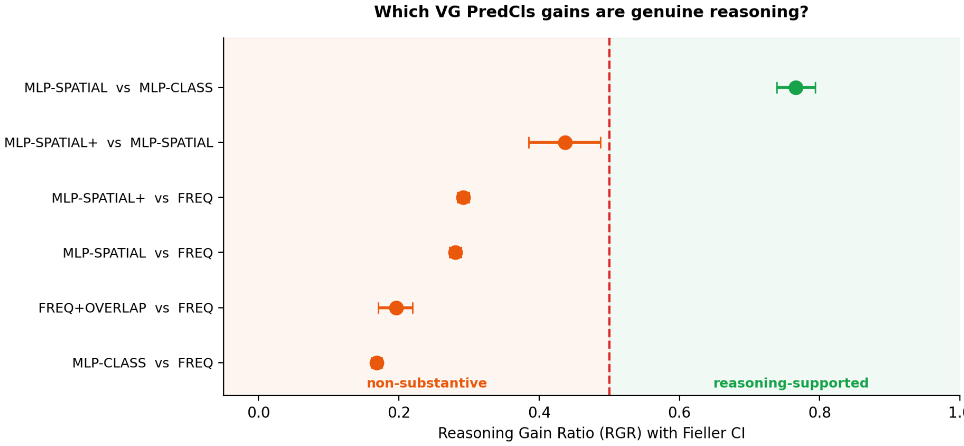

Fig. 10 **Gain attribution.** RGR with Fieller CIs; the dashed line is the 0.5 decision threshold. Only the MLP-CLASS→MLP-SPATIAL step, which adds real box geometry, is reasoning-supported.

prior channel more than the small reasoning channel; we therefore report the conservative default $c = \log K$ throughout and recommend it. The stability of the verdict under ϕ coarsening is the most important check: if a relevant frequency channel were missing from ϕ, coarsening it would move the verdict, and it does not.

5 Discussion, Limitations, and Future Work

What WAGER does and does not claim.

WAGER certifies and quantifies the *usable* reasoning a given model family extracts beyond an annotation-frequency prior, and attributes a gain between two models with an anytime-valid interval. It is the right currency for gain attribution, but $\hat{I}^{\text{reason}}$ is family-relative [22]: it measures what *this* model extracts, not an information-theoretic limit, and a non-substantive verdict is a statement about a particular pair of forecasters, not a claim that the task is unsolvable. Equally, a "reasoning-supported" verdict certifies information beyond ϕ, not correctness or robustness in any broader sense.

Fig. 11 Verdict stability. RGR (MLP-SPATIAL vs FREQ) across design choices; dashed line is the 0.5 threshold. The verdict is stable except under the most aggressive clip $c = \log K/2$.

Relation to the benchmark-reliability crisis.

A growing literature argues that static leaderboards overstate capability, whether through frequency shortcuts, data contamination, or memorization, and especially so for multimodal models [19–21]. Most remedies refresh or randomize the benchmark. WAGER is complementary: it keeps the benchmark fixed and instead audits a *reported gain* against an explicit confound, returning a certificate rather than a fresh test set. The two approaches compose—a contamination-resistant benchmark can still be analyzed with WAGER to separate reasoning from prior-fitting.

Threats to validity.

(i) Exchangeability. We assume exchangeability rather than i.i.d. and enforce it with image-blocked permutations, which also handle within-image structural leakage from multiple pairs per image. *(ii) Estimated projection.* Exact validity uses a $\bar{q}_f$ frozen on an independent fold; cell sparsity (Table 1) is handled by KT smoothing and hierarchical backoff, and the Jensen correction debiases the log-projection. *(iii) ϕ must capture the confound.* If a relevant frequency channel is omitted from ϕ, WAGER can over-credit reasoning; we therefore report the most conservative (finest) ϕ and confirm verdict stability under coarsening (Section 4.8). *(iv) Clipping.* The clip c trades robustness for sensitivity; the verdict is stable at the default $c = \log K$ and only flips under aggressive clipping, which we avoid. *(v) Family-relativity.* Because the measured quantity depends on the model family, WAGER compares forecasters, not architectures in the abstract; the cached-logit interface makes the comparison reproducible and exact.

Future work.

The cached-logit interface makes scaling out almost free; each new study requires only a one-time logit dump and a choice of ϕ (Table 2). Three directions follow naturally. *(a) Released SGG checkpoints.* Applying the identical pipeline to MOTIFS, VCTree, and the causal +TDE variant would test whether published scene-graph SOTA survives the RGR test, and whether TDE raises RGR as the causal account predicts. *(b) VQA and multimodal LLMs.* With ϕ the question type, WAGER can audit whether modern vision-language gains over classical baselines reflect visual reasoning or language-prior fitting, where the confound is well documented [4, 5, 17]. *(c) Long-tail detection.* With ϕ the frequency tier, the head/body/tail analysis of Section 4.6 extends directly to LVIS-style rare-category evaluation [6, 18]. In each case the contribution of this paper—the self-prior projection, the one-process certify-and-quantify result, and the certified RGR—carries over unchanged.

6 Conclusion

We introduced WAGER, a distribution-free, anytime-valid framework that decomposes a computer-vision benchmark gain into a prior-recoverable part and a genuine-reasoning part with a finite-sample certificate. Its technical core is the self-prior projection—comparing a model to its own frequency-collapsed self—which makes the null exact without knowing the true prior and the measurement provably calibration-invariant. A single testing-by-betting capital process built on the projection both certifies (an e-value, via Ville's inequality) and quantifies (a growth rate equal to usable reasoning information) reasoning beyond annotation priors, and two such processes combine into the Reasoning Gain Ratio with an anytime-valid interval and an operational decision rule. On simulations WAGER controls type-I error, attains nominal coverage, and recovers injected reasoning with Pearson $r = 0.998$; on a real Visual Genome study over 7.6×10^5 relations it certifies the frequency baseline as non-reasoning and shows that every accuracy gain over it is statistically indistinguishable from improved annotation-frequency-prior fitting, while still recognizing a genuine geometry-driven gain when one is present. By casting benchmark evaluation as a game-theoretic e-process, WAGER turns "is this gain real?" into a reproducible, certified measurement. The framework operates on cached probability vectors and instantiates across SGG, VQA, and LVIS through a single design choice, the prior feature ϕ.

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Appendix A Full proofs

A.1 Theorem 1 (anytime-valid certificate)

Let $\mathcal{F}_i = \sigma(\tilde{d}_1, \ldots, \tilde{d}_i)$ and let λ_i be $\mathcal{F}_{i-1}$-measurable with $0 \le \lambda_i < 1/c$. Because $\tilde{d}_i \in [-c, c]$ we have $1 + \lambda_i \tilde{d}_i \ge 1 - \lambda_i c > 0$, so $W_n \ge 0$. Under H_0, $\mathbb{E}[\tilde{d}_i \mid \mathcal{F}_{i-1}] \le 0$, hence

$$\mathbb{E}[W_i \mid \mathcal{F}_{i-1}] = W_{i-1}\big(1 + \lambda_i\, \mathbb{E}[\tilde{d}_i \mid \mathcal{F}_{i-1}]\big) \le W_{i-1},$$

so (W_n) is a non-negative supermartingale with $\mathbb{E}[W_n] \le W_0 = 1$. Ville's inequality [41] for non-negative supermartingales states $\Pr[\sup_n W_n \ge 1/\alpha] \le \mathbb{E}[W_0]\,\alpha = \alpha$, which is the claimed time-uniform type-I bound; stopping the first time $W_n \ge 1/\alpha$

therefore rejects at level α regardless of the (possibly data-dependent) stopping rule. $\square$

A.2 Theorem 2 (growth rate equals usable reasoning information)

Write $g_i = \log(1 + \lambda_i \tilde{d}_i)$ so $\frac{1}{n} \log W_n = \frac{1}{n} \sum_i g_i$. For the log-optimal predictable bet $\lambda_i^\star = \arg\max_\lambda \mathbb{E}[\log(1 + \lambda \tilde{d}) \mid \mathcal{F}_{i-1}]$, the Kelly/Cover–Thomas theory of horse-race and forecasting games [37–39] gives that the optimal expected log-growth equals the expected log-score differential of the two forecasters. Under exchangeability the strong law yields $\frac{1}{n} \sum_i \tilde{d}_i \to \mathbb{E}[\tilde{d}]$ a.s., and the optimized growth rate converges to the same limit. Taking $c \to \infty$ replaces $\tilde{d}$ by $d = \log q_f(Y \mid X) - \log \bar{q}_f(Y \mid \phi(X))$, so $G(f) = \mathbb{E}[\log q_f(Y \mid X)] - \mathbb{E}[\log \bar{q}_f(Y \mid \phi(X))] =: \mathcal{I}_f^{\text{reason}}$. Non-negativity follows because $\bar{q}_f$ is the ϕ-measurable projection of q_f and Gibbs' inequality makes the conditional log-loss of the coarser forecaster at least that of the finer one. If $q_f = p(\cdot \mid x)$ and $\bar{q}_f = p(\cdot \mid \phi)$, then $\mathcal{I}_f^{\text{reason}} = \mathbb{E} \log \frac{p(Y|X)}{p(Y|\phi(X))} = I(Y; X \mid \phi(X))$ by the chain rule $I(Y; X) = I(Y; \phi(X)) + I(Y; X \mid \phi(X))$, valid since ϕ is a function of X. $\square$

A.3 Corollary 1 (calibration invariance)

Let T_ϕ be any map applied uniformly to all predictions in cell ϕ (e.g. per-cell temperature scaling), and let $q_f^T(\cdot \mid x) = T_{\phi(x)}(q_f(\cdot \mid x))$. Its projection is $\bar{q}_f^T(\cdot \mid \phi) = T_\phi(\bar{q}_f(\cdot \mid \phi))$ because the projection averages within the cell where T_ϕ is constant in form. In the difference $d^T = \log q_f^T(y \mid x) - \log \bar{q}_f^T(y \mid \phi)$ the common transformation cancels up to the within-cell-constant Jacobian term, leaving $\mathbb{E}[d^T] = \mathbb{E}[d]$, so $\mathcal{I}_f^{\text{reason}}$ is invariant. Moreover, for an *arbitrary* miscalibrated q_f, Gibbs' inequality gives $\mathbb{E}[\log q_f(Y \mid X)] \leq \mathbb{E}[\log p(Y \mid X)]$ with the analogous bound for $\bar{q}_f$, so miscalibration can only shrink the measured reasoning, never inflate it above the true value. $\square$

A.4 Theorem 3 (anytime-valid CI for RGR)

By [31], the hedged capital process gives an anytime-valid $(1-\alpha)$ confidence sequence for the mean of a bounded variable; applied to the clipped numerator score $\Delta d = d_{f'} - d_f$ and denominator score $\Delta(d+e) = (d_{f'} + e_{f'}) - (d_f + e_f)$ it yields intervals $\mathcal{N}_{1-\alpha}$ and $\mathcal{D}_{1-\alpha}$ valid simultaneously over n. By a union bound the event $\{\mathbb{E}[\Delta d] \in \mathcal{N}\} \cap \{\mathbb{E}[\Delta(d+e)] \in \mathcal{D}\}$ has probability $\geq 1 - 2\alpha$. On that event Fieller's construction [42] maps the numerator/denominator interval pair to a set containing $\text{RGR} = \mathbb{E}[\Delta d]/\mathbb{E}[\Delta(d+e)]$; the set is a bounded interval when the denominator interval excludes 0 and is the whole line otherwise. The result is a $(1-2\alpha)$ anytime-valid interval for RGR. $\square$

Appendix B Implementation details

Online-Newton-Step (aGRAPA) bet.

With most recent observation z, previous fraction λ, and accumulated curvature A (initialized $A_0 = 1$), the predictable update of [31] is

$$g = \frac{z}{1+\lambda z}, \quad A \leftarrow A + g^2, \quad \lambda \leftarrow \mathrm{clip}\Big(\lambda + \tfrac{2}{2-\ln 3}\, \tfrac{g}{A},\, 0,\, \tfrac{1}{2c}\Big).$$

The cap at $1/(2c)$ (rather than $1/c$) gives numerical headroom while preserving $1 + \lambda\tilde{d} > 0$.

Hierarchical projection and Jensen correction.

For each cell the shrunk mean toward a parent distribution π is $\hat{\bar{q}} = \frac{\sum_j q_j + m\pi}{n+m}$, with hierarchy cell→coarse (subject)→global. Estimating $\log \bar{q}$ by $\log \hat{\bar{q}}$ is biased downward by Jensen's inequality; we add the second-order delta-method correction $\frac{1}{2}\,\widehat{\mathrm{Var}}[\hat{\bar{q}}]/\hat{\bar{q}}^2$ (capped at $\log K$), where $\widehat{\mathrm{Var}}$ is the within-cell sample variance divided by the cell count. The correction cancels in $d{+}e = I^{\mathrm{tot}}$ and only reallocates mis-credited reasoning into the prior channel.

Anytime-valid CI by hedged capital.

For a candidate mean m we run two predictable capital processes on the centered residuals $x_i - m$, one betting the residual is positive and one negative, and combine them as $W = (W^+ + W^-)/2$, itself a non-negative martingale with $\mathbb{E}[W] \le 1$ under $H_0 : \mathbb{E}[x] = m$. The interval is $\{m : \sup_n W(m) < 1/\alpha\}$. With a predictable fixed λ estimated from a held-out prefix, the whole grid is evaluated by a vectorized cumulative sum, chunked to bound memory.

Hyperparameters.

Table B1 lists the defaults used in all experiments unless ablated.

Table B1 Default hyperparameters.

Symbol	Meaning	Value
c	clip / log-ratio scale	$\log K$
α	error level	0.05
split	projection / eval fold	40% / 60% of test images
m	shrinkage pseudo-count	1.0
R	image-blocked permutations	24
backoff	projection hierarchy	cell → subject → global

Appendix C Reproducibility

Data preparation parses the Visual Genome relationships export into a single array of (subject, object, predicate, boxes, image id, split) tuples under the VG150 vocabulary. Models are trained once on the training split and their predicate distributions on the test split are cached; WAGER then runs on the cache. Object/spatial MLPs are small (two hidden layers, dropout 0.1, AdamW) and train in minutes on a single GPU; all WAGER computation is deterministic CPU NumPy under fixed seeds. The simulation suite, the real study (Tables 3, 5), the stratified analysis (Table 4), and all figures are produced by scripts released with the code, each reading the same cached artifacts so the numbers in the paper and the figures derive from one run.